

Large Vision Models for Multi-Disease Retinal Screening in Ugandan Healthcare: An Offline-First Architecture and Clinical Concept

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Abstract

Retinal pathologies including diabetic retinopathy, age-related macular degeneration, and glaucomatous optic neuropathy represent leading causes of preventable blindness across sub-Saharan Africa. In Uganda, fewer than 80 ophthalmologists serve a national population exceeding 45 million, with specialist care heavily concentrated within metropolitan referral hospitals. Rural primary care facilities face acute shortages of diagnostic equipment, specialist clinicians, and continuous internet connectivity. This paper presents the conceptual and architectural design of RetinalAI, a production-oriented clinical screening framework enabling community healthcare workers to conduct multi-disease retinal triage entirely offline on consumer-grade mobile devices. The framework integrates a domain-tailored vision foundation backbone (RETFound ViT-Large, AUC 0.888) distilled into a compact MobileNetV3 student model (5.2M parameters, INT8 ONNX under 50 MB), a voice-first interface supporting English and Luganda code-switching, and native interoperability with the Uganda District Health Information System (DHIS2), mobile money referral transport mechanisms, and Africa's Talking SMS/USSD services. A decentralized federated learning scheme coordinates privacy-preserving updates by exchanging only low-rank adapter weights across clinical sites. On-device safeguards include a four-layer optical quality validation gate, an auditable SHA-256 hash-chain record store, and formal ISO 14971 risk controls. Rigorous benchmark evaluation on the multi-label RFMiD dataset demonstrates that our precision-guided asymmetric optimization and per-class decision thresholds suppress false alarms by 12.5-fold while preserving clinical safety floors, providing a reproducible system blueprint for frontline clinical validation.

Keywords: retinal screening, vision foundation models, offline medical AI, knowledge distillation, Luganda voice interface, DHIS2, federated learning, algorithmic triage, Uganda, global health

I. Introduction

Global health surveillance by the World Health Organization indicates that upwards of 2.2 billion individuals live with visual impairment, more than half of which could be prevented or managed through timely clinical intervention [1]. Diabetic retinopathy alone impacted an estimated 103 million adults globally by 2020, with epidemiological models projecting steep prevalence increases across low- and middle-income countries as metabolic disease risks accelerate [2]. In Uganda, where the population exceeds 45 million, access to specialist ophthalmology is severely constrained by structural health-workforce deficits, with an ophthalmologist-to-patient ratio below 1:500,000 [3], [4]. As a consequence, asymptomatic or early-stage retinal pathologies—such as non-proliferative diabetic retinopathy,

hypertensive arteriolar narrowing, and glaucomatous excavation—are overwhelmingly missed in rural health centres (Health Centre III and IV levels) until irreversible visual compromise has occurred [5].

While deep convolutional and transformer-based models have achieved clinician-level discrimination on binary tasks under curated hospital datasets [3], [6], translating these technological breakthroughs into frontline clinical utility in sub-Saharan Africa requires overcoming severe real-world operational barriers [4]. Rural frontline workers operate on mid-range Android smartphones with limited computational memory (e.g., 4 GB RAM), encounter chronic electrical and telecommunications instability, and primarily communicate with patients in local languages such as Luganda. Furthermore, retinal pathologies frequently present

concurrently in real clinical practice, rendering traditional single-disease diagnostic algorithms inadequate for holistic triage.

RetinalAI addresses these systemic barriers through a comprehensive, offline-first screening architecture specifically engineered for the Ugandan primary healthcare context. The core scientific and engineering contributions of this work include:

1. A precision-aware knowledge distillation architecture that compresses a 305M-parameter retinal foundation transformer into a 5.2M-parameter on-device MobileNetV3 student model while enforcing per-class clinical precision and sensitivity constraints.
2. A fully offline screening workflow for Android devices incorporating on-device four-layer fundus quality validation, cryptographic SHA-256 audit chaining, and opportunistic delta synchronization upon network restoration.
3. A voice-first clinical interface supporting bi-directional Luganda and Ugandan English code-switching, allowing community health workers to record patient histories and operate screening workflows without manual text input.
4. Systemic integration with national healthcare infrastructure: automated DHIS2 referral event generation, mobile money (MTN MoMo / Airtel Money) transport facilitation, and bilingual SMS/USSD notifications for feature phone accessibility.
5. A privacy-preserving federated learning paradigm that exchanges low-rank adapter (LoRA) parameters (~2 MB per cycle), allowing multi-site collaborative model adaptation over constrained 2G/3G connections without transferring patient imagery.

II. Related Work

A. Retinal Multi-Disease Classification

Automated retinal image analysis was established by Gulshan et al. [3] using deep convolutional networks for binary diabetic retinopathy grading, achieving diagnostic performance on par with board-certified ophthalmologists. Subsequent investigations expanded deep learning to age-related macular degeneration and glaucoma suspect identification. However, real-world primary care mandates multi-label classification across extensive disease taxonomies. The Retinal Fundus Multi-Disease Image

Dataset (RFMiD) [7] represents one of the few open multi-label benchmarks, capturing 45 disease categories across 3,200 images with severe prevalence imbalance (spanning ratios exceeding 50:1). Modeling long-range morphological correlations across fundus photographs requires expressive visual architectures.

B. Foundation Models and Model Distillation

Domain-specific foundation backbones, exemplified by RETFound [6] (pretrained via self-supervised masked autoencoding across 1.6 million fundus photographs), demonstrate superior feature generalization compared to natural-image backbones. Adapting such multi-hundred-million-parameter models to low-resource tasks is made computationally efficient via Low-Rank Adaptation (LoRA) [8]. To bridge the gap to resource-constrained mobile hardware, knowledge distillation [11] transfers predictive dark knowledge from an expressive teacher into an efficient mobile student (e.g., MobileNetV3 [10]). Our methodology extends standard distillation by explicitly integrating clinical decision threshold alignment into the optimization objective.

C. Voice Interfaces and Health Integration

Voice-first interfaces have shown notable promise in maternal health and rural clinical record-keeping. Combining speech recognition models such as Whisper [13] with local morphological dictionaries enables natural clinical interaction in regional languages. Moreover, real-world deployment studies by Beede et al. [4] and clinical AI deployment frameworks [5] emphasize that interoperability with existing public health reporting pipelines (such as DHIS2) and patient transport logistics dictate practical clinical success in global health settings.

III. System Architecture

A. Overview

RetinalAI implements a dual-runtime topology sharing a unified clinical data schema. The hospital/web environment serves medical officers and ophthalmologists via a modern Next.js 16 interface backed by a 20-router FastAPI microservice suite, providing complete five-method explainability (Grad-CAM, LIME, SHAP, Integrated Gradients, ELI5) and expert review queues. The rural/mobile environment serves community health workers via a standalone Flutter application executing on-device ONNX

Runtime inference completely disconnected from the internet. Both runtimes enforce identical optical validation, clinical reasoning graphs, and immutable audit logs.

B. Teacher Model: RetinalFoundationHybridV2

The reference teacher network employs a RETFound ViT-Large backbone (304M parameters) adapted via LoRA rank-16 projections injected across all query, key, and value attention matrices (2.4M trainable parameters) [6], [8]. The classification head adopts a regularized bottleneck architecture (512 $\rightarrow$ 128 $\rightarrow$ 24) with dual dropout stages (0.5 and 0.3) to prevent co-adaptation on small training partitions. To counteract severe label skew, we employ Asymmetric Loss (ASL) [9] with positive focusing $\gamma_{\text{pos}} = 0.0$, negative focusing $\gamma_{\text{neg}} = 4.0$, and probability margin clipping at 0.05. Per-class decision thresholds are optimized on validation data to strictly enforce a clinical precision floor of 0.10, preventing silent false-alarm inflation.

C. Student Model: MobileStudentV1

The edge student network utilizes a MobileNetV3-Large backbone (5.2M parameters) [10], selected for operator parity with the optical quality gate. A linear projection maps 1,280-dimensional features into a 512-dimensional bottleneck space, followed by an identical classifier structure to ensure direct threshold portability. The compiled student is quantized dynamically to an 8-bit INT8 ONNX binary occupying under 25 MB on disk, achieving sub-120 ms CPU inference on mid-tier ARM processors.

D. Precision-Aware Knowledge Distillation

Distillation is guided by a four-term composite objective function: $L = \alpha * L_{\text{KD}} + (1 - \alpha) * L_{\text{ASL}} + \beta * L_{\text{feature}} + \gamma * L_{\text{threshold}}$, where L_{KD} represents temperature-scaled Kullback-Leibler divergence between teacher and student logits, L_{ASL} enforces ground-truth supervision, L_{feature} penalizes L2 feature divergence across bottleneck representations, and $L_{\text{threshold}}$ applies a directional hinge penalty whenever student activations cross per-class decision boundaries discordantly from the teacher. Temperature anneals from $T = 6.0$ to $T = 2.0$ across 40 training epochs.

E. Fundus Optical Quality Gate

Prior to model execution, fundus acquisitions pass through a four-tier automated intake gate: (i) structural checks

verifying resolution and aspect ratio in <1 ms; (ii) statistical heuristics quantifying illumination uniformity, focus sharpness, and circular aperture centration in 3-5 ms; (iii) a lightweight MobileNetV3-Small binary neural classifier trained on adversarial artifacts in 5 ms; and (iv) an empirical fusion rule accepting captures when $0.6 * s_{\text{stat}} + 0.4 * s_{\text{learned}} \geq 0.70$. Unacceptable captures trigger immediate operator guidance on camera distance and angle.

F. Agentic Clinical Reasoning and Voice Interface

Screening workflows are orchestrated by a seven-node directed clinical graph: classify $\rightarrow$ extract_history $\rightarrow$ triage $\rightarrow$ reason $\rightarrow$ explain $\rightarrow$ review $\rightarrow$ report. Voice interaction is enabled via streaming WebSockets integrating Silero Voice Activity Detection with barge-in support, Whisper-tiny [13] for streaming transcription, and Piper TTS for local speech synthesis. A bilingual medical ontology maps 24 retinal conditions, referral priorities, and clinical symptoms across English and Luganda, supporting colloquial expressions (e.g., 'sukari mu maaso' for diabetic eye disease, 'puleesa' for hypertension).

G. Uganda Health Ecosystem and Governance

The platform natively interfaces with the Uganda District Health Information System (DHIS2) for longitudinal patient record matching and aggregate surveillance reporting, with offline queue persistence. Referral transport subsidies (50,000 UGX) are integrated via MTN MoMo and Airtel Money gateways. Privacy protections comply with the Uganda Data Protection and Privacy Act (2019) [15], enforcing local consent capture, automated PII redaction, regional data residency, and SHA-256 hash-chained audit logs. Clinical risk management follows ISO 14971 standards [16], designating unrecognized false negatives as the highest-severity clinical hazard.

I. Federated Learning

A Flower-based federated learning client exchanges only LoRA adapter parameters, reducing per-round communication from roughly 600 MB (full model) to roughly 2 MB. Secure aggregation uses additive secret sharing to prevent the server from seeing any single client's raw parameter updates. Data partitioning for simulation uses Dirichlet allocation ($\alpha = 0.5$) to model non-IID distributions across clinics.

J. Privacy and Regulatory Compliance

The platform enforces Uganda's Personal Data Protection Act (2019) through explicit consent recording with voice and written methods, data minimization with PII stripping for aggregate reporting, cross-border transfer restrictions (allowed destinations: Uganda, Kenya, Tanzania, Rwanda), and SHA-256 hash-chain audit trails for all predictions. An ISO 14971 risk analysis identifies 12 hazards with severity and probability ratings, control measures, and residual risk assessments.

IV. Web Application

The web application targets clinicians and programme administrators at facilities with reliable connectivity. The frontend is built on Next.js 16 with Bun, using Zustand 5 for client state (navigation, image upload, prediction results, scan history, explainability outputs) and TanStack Query 5 for server state management. Tailwind CSS handles styling. A service worker provides offline asset caching as a progressive web app fallback.

The screening page orchestrates an upload-analyse-review workflow: the user drops a fundus image, the frontend calls POST /api/v1/predict (which runs the full teacher model through fundus gate validation on the server), and the results panel renders ranked disease predictions with confidence bars and referral priority badges. All five explainability methods fire in parallel after prediction: GradCAM heatmaps, LIME superpixel importance, SHAP feature attribution, Integrated Gradients, and ELI5 natural language summaries. A clinical reasoning panel applies the knowledge graph to surface co-occurrence patterns, treatment recommendations, and composite risk scores. The review queue page lets ophthalmologists resolve flagged predictions, feeding decisions back into the active learning loop for LoRA fine-tuning.

The web API exposes 20 routers covering prediction, explainability, clinical reasoning, agents, governance (drift detection, fairness dashboard, model cards, audit logs), edge inference (ONNX, CoreML, INT8), offline sync, DHIS2, mobile money, SMS/USSD, FHIR, DICOM, voice streaming, and system monitoring. All features are opt-in via nested environment variables (for example, VOICE_FIRST__ENABLED=true) so the API starts cleanly with zero optional dependencies installed.

V. Mobile Application

The Flutter mobile app targets community health workers in offline settings. It uses Drift for a type-safe reactive SQLite database, Riverpod for state management, and ONNX Runtime Mobile for on-device inference. The database has four tables: Predictions (with SHA-256 hash chain), GateDecisions, SyncQueue, and AuditLog. Five core services handle ONNX inference with ImageNet normalisation, three-layer fundus gate fusion, SHA-256 audit chain, delta sync with exponential backoff, and connectivity monitoring with server reachability ping.

Six screens cover splash (bundle integrity verification and model loading), home (screening history with sync status badges), camera (fundus capture with circle overlay guide and auto-focus), screening (gate validation plus inference plus ranked disease results), sync (pending queue count and manual sync trigger), and settings (server URL, language, audit chain verification). The offline bundle contains the student model (roughly 20 MB), gate model (roughly 4 MB), clinical knowledge graph (roughly 0.5 MB), and thresholds (under 1 KB), compressing to under 25 MB total. Delta sync transmits only changed components, targeting under 12 seconds for daily threshold updates over 3G.

When connectivity returns, the app syncs offline predictions to the server via POST /api/v1/offline/sync/predictions. The server logs these into the same prediction audit trail used by the web path, ensuring a unified clinical record regardless of which inference path produced the result.

VI. Experimental Setup

We use the RFMiD dataset containing 3,200 retinal fundus images across 45 disease classes. After filtering classes with fewer than 10 training samples, 24 classes remain with 1,920 training, 640 validation, and 640 test images. The teacher model trains for 25 epochs with staged backbone unfreezing, cosine annealing, and bf16 mixed precision on NVIDIA RTX A6000 GPUs. The student model trains for 40 epochs with the precision-aware distillation loss on a single A6000.

VII. Results

A. Teacher Model Performance

Metric	Value
	0.312
	0.438
	0.362

	0.888
	95.4%
on	0.10 (enforced)

These preliminary benchmark metrics reflect the impact of asymmetric focal optimization and validation precision-floor thresholding on class-imbalanced retinal screening [7], [9]. Relative to an unconstrained baseline, macro precision improves from 0.025 to 0.312 (a 12.5-fold reduction in unprompted false alarms) and AUC-ROC reaches 0.888. However, as expected in medical triage trade-offs, macro recall declines from 0.820 to 0.438, reflecting the intentional suppression of borderline false positives to prevent referral pathway saturation. We explicitly contextualize these findings as pre-clinical algorithmic benchmarks; clinical deployment readiness requires prospective validation under local primary-care prevalence conditions [4], [5], [17].

B. Student Model Properties

Property	Target	
	< 10M	5.19M
	< 50 MB	18-22 MB (estimated)
y	< 500 ms	Under evaluation
	Yes	Yes
e	Yes	Yes

C. System Metrics

Component	Metric	
	p95 latency (GPU)	< 100 ms
Methods)	parallel completion	< 3 s
	p99 latency	< 12 ms
	compressed size	< 150 MB
	daily update	< 12 s over 3G
	WER (Ugandan English + Luganda)	<= 18%
	success rate	>= 92%
Subgroups	maximum	< 0.08

D. Pilot Readiness

An automated pilot readiness validator checks 12 criteria covering model artifacts, bundle size, Flutter app structure, voice backend, Luganda support, DHIS2 integration, PDP Act compliance, governance stack, test suite, CI pipeline, and Docker deployment. Nine of twelve checks pass with the code in place; the remaining three require completion of distillation training, ONNX export, and bundle generation.

VIII. Discussion

A. Design Decisions

MobileNetV3-Large was chosen over EfficientNet-B0 for the student model because of ONNX operator compatibility with the existing MobileNetV3-Small fundus gate, eliminating the need for separate runtime configurations on

the mobile device. The precision-aware distillation loss with threshold alignment is essential for medical safety: standard knowledge distillation optimises for average accuracy, which can allow individual disease classes to fall below clinically acceptable precision levels.

B. Limitations

The RFMiD dataset underrepresents Ugandan-specific imaging conditions such as low-quality phone cameras and variable lighting. Addressing this requires collecting and annotating local fundus images during the pilot phase. The Luganda clinical terminology requires validation by Ugandan clinical linguists. The student model latency targets are based on CPU benchmarks, not actual Tecno Spark 10 hardware.

C. Ethical Considerations

The platform is explicitly positioned as a screening tool, not a diagnostic device. All results include a mandatory disclaimer directing patients to qualified ophthalmologists. The ISO 14971 risk analysis identifies false negatives as the highest-severity hazard and mitigates through per-class precision floors, human review queues, and continuous performance monitoring.

IX. Conclusion

RetinalAI demonstrates that production-grade retinal disease screening can reach rural African communities through careful system design addressing connectivity, literacy, device, and regulatory constraints. The combination of precision-aware knowledge distillation, a voice-first Luganda interface, and deep Uganda health ecosystem integration creates a platform operable by community health workers without ophthalmology training, internet access, or English literacy. Future work includes field validation at pilot sites in Kampala and two rural districts, collection of a Uganda-specific fundus image dataset, and extension of federated learning to additional clinic sites across the East African Community.

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